HUMPHREY VIRTUAL FARMS LLC // WP-3: SITE PLC, 60HZ SCADA, & SAE J1939 CAN-BUS ICD
Entity: Humphrey Virtual Farms LLC (‘CAGE: 1AHA8‘ | ‘SAM.gov UEI: S1M4ENLHTDH5‘)
Controlling Authority: Jeffery Humphrey, Founder, CEO, & Principal Sovereign Architect
Plug-In Module: ‘CLIN 0029 (WP-3 Site Interface Adapter)‘ | **Enclave Tier:** ‘PUBLIC‘
Host Ebony Baseline Merkle Root: ‘cc1ee1c8d29b2ae5e856fc0fcb672a52e7e49d50369499febe14e1
8e9150949a‘

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1. 60Hz Microgrid Substation Relay Binding (‘CLIN 0003‘ -- IEEE C37.118 / Modbus / DNP3) Tj T* ( )
* **Measured Bus Frequency:** ‘59.985 Hz‘ (Deviation: ‘0.0150 Hz‘ | Limit: ‘<= 0.50 Hz‘)
* **Rate-of-Change-of-Frequency (RoCoF):** ‘-0.12 Hz/s‘ (Limit: ‘<= 1.50 Hz/s‘)
* **PCC Islanding Breaker Trip State:** ‘False‘
* **Hardware Register Map:** ‘[REDACTED_MODBUS_DNP3_REGISTER_MAP_DFARS_252_227_7018]‘

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2. 480V 3-Phase Grain Silo Contactor & Plenum PLC Binding (‘CLIN 0009‘ -- NEMA MG-1)
* **3-Phase RMS Voltage Average:** ‘480.00 V AC‘
* **NEMA MG-1 Phase Voltage Unbalance:** ‘0.2500%‘ (Limit: ‘<= 2.00%‘)
* **Contactor Interlock Safe:** ‘True‘ | **Command:** ‘ENERGIZE_480V_CONTACTOR‘
* **PLC Coil & Register Map:** ‘[REDACTED_480V_PLC_COIL_MAP_DFARS_252_227_7018]‘

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3. SAE J1939 29-Bit Tactical CAN-Bus Decoder (‘CLIN 0008‘ -- PGN 61444 EEC1)
* **29-Bit Extended CAN Identifier:** ‘0x0CF00400‘ (‘Priority: 3‘ | ‘PGN: 61444‘ | ‘Source:
0‘)
* **Decoded Engine Speed (SPN 190):** **‘1800.00 RPM‘**
* **CAN-Bus Physical & Scaling Binding:** ‘[REDACTED_J1939_SPN_SCALING_DFARS_252_227_7018]‘

WORM WP-3 Interface Control Document Seal (SHA-256): ‘3a7f92d1f670ff1dbdc9e4259d1b60e0
fa14c54e46c1ed37ad484fbcb042a343‘